# Bingjie YAN

Medical Agents · Multimodal Ophthalmic AI · Federated Learning

*The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong SAR, China*

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*"Keep the curiosity."*

## EDUCATION

### School of Optometry, The Hong Kong Polytechnic University — *Hong Kong, China*

Doctor of Philosophy — *2025.09 - Present*

- **Supervisor:** Prof. Danli SHI; **Co-supervisor:** Prof. Mingguang HE.
- **Research Topic:** Towards a Generalizable Multimodal Tool-Use Agent for Ophthalmology.

### Institute of Computing Technology, Chinese Academy of Sciences (ICT, CAS & UCAS) — *Beijing, China*

Master of Engineering, Electronic and Information Engineering — *2022.09 - 2025.06*

- **Supervisors:** Prof. Yiqiang CHEN and Prof. Xinlong JIANG. **GPA:** 3.79/4.
- **Research Topic:** Heterogeneous federated learning and foundation models for privacy-preserving healthcare.
- **Awards:** **National Scholarship (3%)**, Merit Student Awards (10%), The First Prize Scholarship (20%).

### School of Computer Science and Technology, Hainan University (211) — *Hainan, China*

Bachelor of Engineering, Software Engineering for Big Data — *2018.09 - 2022.06*

- **GPA:** 3.68/4 (89.65/100). **Ranking:** 10/179.
- **Awards:** The First Prize Scholarship (3%), Outstanding Graduate Awards (3%) and Merit Student Awards.

## RESEARCH EXPERIENCE

### Ophthalmic AI Agents | Ph.D. Student

School of Optometry, The Hong Kong Polytechnic University — *2025.09 - Present*

- Co-developed EyeAgent, integrating 54 ophthalmic tools and 32 clinical skills for multimodal decision support (co-first author; NEJM AI, accepted). [arXiv] [EyeAgent Code] [EyeTools Code]
- Developed EyeFlow, a diagnostic multi-agent framework combining image analysis, specialist debate, and retrieval (first-author abstract, APTOS 2026). [Abstract]

### Privacy-Preserving Federated Learning for Healthcare | M.Eng. Student

Institute of Computing Technology, Chinese Academy of Sciences — *2022.09 - 2025.06*

- Designed FedEYE with Aier Eye Hospital for privacy-preserving ophthalmic collaboration across 50+ institutions (co-first author; Patterns 2024). [Paper] [Code]
- Developed Buffalo, using cross-modal prototypes and federated foundation model collaboration to address missing modalities in biomedical vision-language learning (first author; CIKM 2024). [Paper] [Code]
- Proposed AFL-CS for asynchronous federated learning using cosine similarity to address client heterogeneity (ICPADS 2023). [Paper] [Code]
- Contributed to Model Trip, a privacy- and fairness-aware model-fusion method across federations (ICDE 2024). [Paper]

### Large and Small Model Collaboration | Research Intern

Institute of AI Industry Research, Tsinghua University — *2024.07 - 2024.10*

- Surveyed knowledge transfer and inference-time collaboration between large and small models under privacy and resource constraints.
- Co-authored *Towards Harnessing the Collaborative Power of Large and Small Models for Domain Tasks.* [arXiv] [Resources]

### Constrained Online Optimization | Research Assistant

Hong Kong Baptist University — *2024.06 - 2024.07*

- Analyzed dynamic regret and hard constraint violations for time-varying online optimization using Lyapunov drift.
- Implemented COLDQ and baselines for online optimization; co-authored the AAAI 2025 paper. [Paper]

## WORK EXPERIENCE

### FedML Inc.

Remote Research Intern — *2022.06 - 2022.09*

- Enhanced FedCV by integrating YOLOv5 object detection into its federated computer vision framework.
- Adapted the FLamby benchmark's seven real-world healthcare datasets to the FedML platform.

## PUBLICATIONS

Full publication list: Homepage | Google Scholar (800+ citations). * Equal contribution; † corresponding author.

Selected Publications

[1] **FedEYE: A Scalable and Flexible End-to-end Federated Learning Platform for Ophthalmology**
**Bingjie YAN***, Danmin CAO*, Xinlong JIANG, Yiqiang CHEN†, Weiwei DAI†, Fan DONG, Wuliang HUANG, Teng ZHANG, Chenlong GAO, Qian CHEN, Zhen YAN, Zhirui WANG
*Patterns, Cell Press, JCR-Q1, 2024 (IF: 7.4)* [Paper] [Code]

[2] **EyeAgent: An Agentic AI System for Multimodal Clinical Decision Support in Ophthalmology**
Danli SHI*, Xiaolan CHEN*, **Bingjie YAN***, Weiyi ZHANG, Jiancheng YANG, Ruoyu CHEN, Xinyuan WU, Siyu HUANG, Bowen LIU, Ziyu GAO, Yue WU, Pusheng XU, Senlin LIN, Kai JIN, Xia GONG, Yih Chung THAM, Xiujuan ZHANG, Li DONG, Yuzhou ZHANG, Jason YAM, Guangming JIN, Xiaohu DING, Haidong ZOU, Yalin ZHENG, Zongyuan GE, Mingguang HE†
*NEJM AI, accepted* [arXiv] [EyeAgent Code] [EyeTools Code]

[3] **Model Trip: Enhancing Privacy and Fairness in Model Fusion across Multi-Federations for Trustworthy Global Healthcare**
Qian CHEN, Yiqiang CHEN†, **Bingjie YAN**, Xinlong JIANG, Xiaojin ZHANG, Yan KANG, Teng ZHANG, Wuliang HUANG, Chenlong GAO, Lixin FAN, Qiang YANG
*ICDE'24, CCF-A, 2024, Oral, Acceptance Rate: 25.4%* [Paper]

Medical Agents and Agentic AI

[4] **EyeFlow: A Self-Evolving Multi-Agent Framework for Knowledge-Grounded Ophthalmic Diagnosis**
**Bingjie YAN**, Xiaolan CHEN, Mingguang HE, Danli SHI
*11th Asia Pacific Tele-Ophthalmology Society (APTOS) Symposium 2026, Free Paper Abstract* [Abstract]

[5] **Empowering Locally Deployable Medical Agent via State Enhanced Logical Skills for FHIR-based Clinical Tasks**
Wanrong YANG, Zhengliang LIU, Yuan LI, **Bingjie YAN**, Lingfang LI, Mingguang HE, Dominik WOJTCZAK, Yalin ZHENG, Danli SHI
*arXiv:2603.06902, 2026* [arXiv]

[6] **From visual question answering to intelligent AI agents in ophthalmology**
Xiaolan CHEN, Ruoyu CHEN, Pusheng XU, Xiaojie WAN, Weiyi ZHANG, **Bingjie YAN**, Xianwen SHANG, Mingguang HE, Danli SHI
*British Journal of Ophthalmology, 110(1), 1–7, JCR-Q1, 2026, IF (2025): 4.5* [Paper]

Multimodal and Biomedical AI

[7] **EyeWorld: A Generative World Model of Ocular State and Dynamics**
Ziyu GAO, Xinyuan WU, Xiaolan CHEN, Zhuoran LIU, Ruoyu CHEN, Bowen LIU, **Bingjie YAN**, Zhenhan WANG, Kai JIN, Jiancheng YANG, Yih Chung THAM, Mingguang HE, Danli SHI
*arXiv:2603.14039, 2026* [arXiv]

[8] **Benchmarking large multimodal models for ophthalmic visual question answering with OphthalWeChat**
Pusheng XU, Xia GONG, Xiaolan CHEN, Weiyi ZHANG, Jiancheng YANG, **Bingjie YAN**, Meng YUAN, Yalin ZHENG, Mingguang HE, Danli SHI
*Advances in Ophthalmology Practice and Research, 6(1), 33–41, JCR-Q1, 2026, IF (2025): 4.5* [Paper]

[9] **EyeGraphGPT: Knowledge Graph Enhanced Multimodal Large Language Model for Ophthalmic Report Generation**
Zhirui WANG, Xinlong JIANG†, Chenlong GAO, Fan DONG, Weiwei DAI, Bingyu WANG, **Bingjie YAN**, Qian CHEN, Wuliang HUANG, Teng ZHANG, Yiqiang CHEN
*IEEE BIBM'24, CCF-B, 2024, Oral* [Paper]

[10] **Correlation-Driven Multi-Modality Graph Decomposition for Cross-Subject Emotion Recognition**
Wuliang HUANG, Yiqiang CHEN†, Xinlong JIANG, Chenlong GAO, Qian CHEN, Teng ZHANG, **Bingjie YAN**, Yifan WANG, Jianrong YANG
*ACM MM'24, CCF-A, CORE-A*, 2024, Poster, Acceptance Rate: 26.2%* [Paper]

[11] **An Improved Method for the Fitting and Prediction of the Number of COVID-19 Confirmed Cases Based on LSTM**
**Bingjie YAN**, Jun WANG, Zhen ZHANG, Xiangyan TANG†, Yize ZHOU, Guopeng ZHENG, Qi ZOU, Yao LU, Boyi LIU, Wenxuan TU, Neal XIONG
*Computers, Materials and Continua, SCI, JCR-Q2, 2020 (IF: 3.772)* [Paper]

Federated Learning and Privacy Preserving

[12] **FairFHTL: Achieving Task-Agnostic Fairness in Federated Hetero-Task Learning**
Teng ZHANG, Yiqiang CHEN, Xinlong JIANG, Wuliang HUANG, Qian CHEN, Chenlong GAO, Zhirui WANG, **Bingjie YAN**
*IEEE International Conference on Multimedia and Expo (ICME 2025), pp. 1–6, CCF-B, Acceptance Rate: 27.3%* [DOI]

[13] **Towards Harnessing the Collaborative Power of Large and Small Models for Domain Tasks**
Yang LIU, Kejia ZHANG, **Bingjie YAN**, Tianyuan ZOU, Jianqing ZHANG, Zixuan GU, Xiangsen CHEN, Jianbing DING, Xidong WANG, Jingyi LI, Xiaozhou YE, Ye OUYANG, Qiang YANG, Ya-Qin ZHANG
*arXiv:2504.17421, 2025* [arXiv] [Resources]

[14] **Buffalo: Biomedical Vision-Language Understanding with Cross-Modal Prototype and Federated Foundation Model Collaboration**
**Bingjie YAN**, Qian CHEN, Yiqiang CHEN†, Xinlong JIANG, Wuliang HUANG, Bingyu WANG, Zhirui WANG, Chenlong GAO, Teng ZHANG
*ACM CIKM'24, CCF-B, CORE-A, 2024, Oral, Acceptance Rate: 23.2%* [Paper] [Code]

[15] **PrivFusion: Privacy-Preserving Model Fusion via Decentralized Federated Graph Matching**
Qian CHEN, Yiqiang CHEN†, Xinlong JIANG, Teng ZHANG, Weiwei DAI, Wuliang HUANG, **Bingjie YAN**, Zhen YAN, Wang LU, Bo YE
*IEEE Transactions on Knowledge and Data Engineering (TKDE), CCF-A, CORE-A*, JCR-Q1, 2024 (IF: 10.4)* [Paper]

[16] **FedTAM: Decentralized Federated Learning with a Feature Attention Based Multi-teacher Knowledge Distillation for Healthcare**
Tingting MOU, Xinlong JIANG†, Jie LI, **Bingjie YAN**, Qian CHEN, Teng ZHANG, Wuliang HUANG, Chenlong GAO, Yiqiang CHEN
*IEEE ICPADS, CCF-C, 2023, Oral, Acceptance Rate: 19.4%* [Paper]

[17] **FedCM: A Real-time Contribution Measurement Method for Participants in Federated Learning**
**Bingjie YAN**, Boyi LIU†, Lujia WANG, Yize ZHOU, Zhixuan LIANG, Ming LIU, Cheng-Zhong XU
*IJCNN, CCF-C, 2021* [Paper] [Code]

[18] **Experiments of Federated Learning for COVID-19 Chest X-ray Images**
**Bingjie YAN**, Jun WANG, Jieren CHENG†, Yize ZHOU, Yixian ZHANG, Yifan YANG, Li LIU, Haojiang ZHAO, Chunjuan WANG, Boyi LIU
*ICAIS, EI, 2021, Acceptance Rate: 30%* [Paper]

Large Language Model & Knowledge Distillation

[19] **Survey on Knowledge Distillation for Large Language Models: Methods, Evaluation, and Application**
Chuanpeng YANG, Yao ZHU, Wang LU, Yidong WANG, Qian CHEN, Chenlong GAO, **Bingjie YAN**, Yiqiang CHEN
*ACM Transactions on Intelligent Systems and Technology, 16(6), Article 143, CCF-C, JCR-Q1, 2025 (IF: 10.7)* [Paper]

Online Optimization

[20] **Doubly-Bounded Queue for Constrained Online Learning: Keeping Pace with Dynamics of Both Loss and Constraint**
Juncheng WANG, **Bingjie YAN**, Yituo LIU
*AAAI, 39(20), 21135–21143, CCF-A, 2025, Acceptance Rate: 23.4%* [Paper]

[21] **AFL-CS: Asynchronous Federated Learning with Cosine Similarity-based Penalty Term and Aggregation**
**Bingjie YAN**, Xinlong JIANG†, Yiqiang CHEN†, Chenlong GAO, Xuequn LIU
*IEEE ICPADS, CCF-C, 2023, Oral* [Paper] [Code]

Computing Systems

[22] **Im2col-Winograd: An Efficient and Flexible Fused-Winograd Convolution for NHWC Format on GPUs**
Zhiyi ZHANG, Pengfei ZHANG, Zhuopin XU, **Bingjie YAN**, Qi WANG†
*ICPP'24, CCF-B, 2024, Oral, Acceptance Rate: 29.4%* [Paper] [Code]

[23] **UbiMeta: A Ubiquitous Operating System Model for Metaverse**
Yiqiang CHEN†, Wuliang HUANG, Xinlong JIANG, Teng ZHANG, Yi WANG, **Bingjie YAN**, Zhirui WANG, Qian CHEN, Yunbing XING, Dong LI, Guodong LONG
*International Journal of Crowd Science (IJCS), 2023* [Paper]

## SELECTED AWARDS

*2025* **Scholarship**, Institute of Computing Technology Ziqiang International Exchange Scholarship
*2024* **National Scholarship (3%)**, Institute of Computing Technology, Chinese Academy of Sciences
*2024* **First-Class Scholarship**, Institute of Computing Technology, Chinese Academy of Sciences
*2022* **Outstanding Graduate**, Hainan University
*2019* **First-Class Scholarship**, Hainan University
*2017* **Silver**, (Intl.) 11th Asia and Pacific Informatics Olympiad, APIO — *Beijing*
*2019* **1st Prize**, (Natl.) The 3rd Silk Road Robotics Innovations Competition — *Xi'an*
*2020* **2nd Prize**, (Natl.) Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM) — *Beijing*
*2020* **2nd Prize**, (Natl.) China Collegiate Computing Contest - Group Programming Ladder Tournament — *China*
*2020* **2nd Prize**, (Natl.) Chinese Collegiate Computing Competition — *Beijing*
*2023* **Silver**, (Natl.) China International College Students' "Internet+" Innovation and Entrepreneurship Competition — *Beijing*
*2020* **Gold & Silver**, (Prov.) The 6th "Internet+" Innovation and Entrepreneurship Competition in Hainan — *Hainan*

## PROFESSIONAL ACTIVITIES

**Conference Reviewer**

- Served as a reviewer for MICCAI (2026), AAAI (2025–2027), and ICME (2025).

**Teaching Assistant**

- *SO1D02: AI in Healthcare*, The Hong Kong Polytechnic University (Semester 1, 2026/27).

**Leadership**

- President, IEEE Hainan University Branch (Mar. 2021–Jun. 2022).
- Co-founder and Vice President, Association of Robotics and Artificial Intelligence (Jul. 2020–Jun. 2022).
- Vice President, Cyberspace Security Association at Hainan University (Sep. 2020–Jun. 2021).

## SKILLS & INTERESTS

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|---|---|
| **Programming** | Python, C/C++, Java, JavaScript, TypeScript, HTML |
| **ML & Training** | PyTorch, TensorFlow, Hugging Face Transformers, PEFT (LoRA), timm, DeepSpeed, verl (RL) |
| **Agents & Serving** | LangChain, LangGraph, MCP, RAG, vLLM, SGLang, Ollama |
| **Coding Agents** | OpenAI Codex, Claude Code, Pi, DeepSeek Harness |
| **Tools & Data** | FastAPI, Gradio, React, Git, Docker, Linux, NumPy, pandas |
| **Language** | Chinese (Native), English (Fluent), IELTS: 6.5 (L:8.5, R:6.5, W:6.0, S:5.5), CET-6: 478, CET-4: 539 |
| **Photography** | Enjoy the life and capture the moments ;) |